Agent-GhibliLock: Deterministic Wavelet-Locked Cross-Modal Style Transfer and ISO/IEC 19566-5 C2PA-Inspired Provenance Compilation for Multimodal Generative Pipelines

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***Abstract*—Enterprise multimodal workflows increasingly chain vision-language models (VLMs) with latent diffusion image and video synthesizers. When an upstream VLM translates a source photograph into natural-language style instructions and a downstream diffusion model synthesizes new pixels (Pixels $\rightarrow$ Prose $\rightarrow$ Pixels), two coupled failure modes arise, which we term *The Captioning Conundrum*: (1) *high-frequency structural phase drift*, where fine-grained corporate emblems, typographic glyphs, and perspective rails are lost across the low-entropy text bottleneck; and (2) *cryptographic provenance severance*, where container-level Coalition for Content Provenance and Authenticity (C2PA) manifests are stripped or invalidated during cross-modal regeneration and social-platform re-encoding. We present Agent-GhibliLock, a deterministic, CPU-executable post-processing compiler that reconciles high-frequency structural preservation, target palette alignment, and dual-layer provenance attestation. First, Agent-GhibliLock decomposes CIELAB luminance (L^*) via an orthonormal K-level 2D Discrete Haar Wavelet Transform (DWT_2), supporting both *global* subband locking and *spatially masked region-of-interest (ROI)* detail locking (weighted by 5×5 Sobel directional coherence so foreground emblems and rails are phase-locked while non-ROI background regions retain smooth painterly gouache brushwork), and proves that arbitrary spatial detail masks $M_k(u, v) \in [0, 1]$ preserve exact spatial mean luminance $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}})$ (Theorem 1). Second, it aligns full 3D (L^*, a^*, b^*) color distributions via Iterative Sliced-Wasserstein Optimal Transport (SWOT) over $D = 3M+3 = 27$ orthonormal axes ($I_3 \cup \{Q_m\}_{m=1}^{M} \subset O(3)$) with deterministic lexicographical tie-breaking, explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma, \tau_\alpha)$ closed-loop controller satisfying an exact deterministic root-mean-square (RMS) Wasserstein perturbation bound (Theorem 2). Third, for Group-of-Pictures (GOP) video synthesis, it suppresses inter-frame flicker via motion-compensated, motion-gated wavelet LL_K exponential moving average (EMA) filtering while committing frame hashes to a binary Merkle tree root. Fourth, it binds hard and soft provenance by injecting a compact C2PA-inspired ISO/IEC 19566-5 JUMBF binary manifest box (`caBX` PNG chunk with IEEE 802.3 CRC-32 and keyed HMAC-SHA256) alongside a blind Quantization-Index-Modulation (QIM) watermark in mid-band 8×8 2D-DCT luminance coefficients supporting both $M_{\text{coord}} = 4$ ($R = \lfloor B/16 \rfloor$) and $M_{\text{coord}} = 5$ odd-repetition ($R = 5B/64$) voting, keyed dither K_{secret}, export self-noise accounting $\sigma^2_{\text{round+clamp}}$, and blind majority-voted coset tamper localization (Theorem 3). Across a stratified $N = 600$ benchmark (evaluating 9 paradigms including a ControlNet $8\times$-VAE adapter baseline, $K \in \{1, 2, 3\}$ wavelet depth ablations, and disaggregated ± 2 LSB vs. lossy JPEG $Q = 85$ attacks), live multi-model Vertex AI Gemini evaluations (`gemini-2.5-flash`, `gemini-2.5-pro`, `gemini-3-flash-preview`), and real natural photographic evaluations (`grace_hopper.jpg` at 64^2–512^2 and `Minduka_Present_Blue_Pack.png`), Agent-GhibliLock achieves $\mathrm{SSIM}_{\text{edge}} = 0.9967$ (0.9977 ROI-masked; 0.9929–0.9966 on real photographs), non-circular $\mathrm{SSIM}_{\text{full}} = 0.9752$, Sobel gradient correlation $\rho_{\text{grad}} = 0.8944$, post-export $\widehat{W}_2^{\text{out}} = 1.9603$ (held-out $\widehat{SW}_2^{\text{eval}} = 2.0739$), 99.9% GOP flicker reduction, and 100.0% tamper detection and blind watermark bit recovery under container stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression.**

***Index Terms*—Cross-Modal Style Transfer, Discrete Haar Wavelet Transform, Sliced-Wasserstein Optimal Transport, C2PA JUMBF Provenance, Quantization Index Modulation, 2D-DCT Watermarking, Multimodal Agentic Pipelines.**

I. Introduction

Modern enterprise creative and media workflows frequently orchestrate multi-stage generative pipelines in which a vision-language model (VLM) inspects an input asset I_{src} and emits structured scene descriptions and stylistic prompts for a downstream latent diffusion image or video synthesizer [1]–[3]. While this modular Pixels $\rightarrow$ Prose $\rightarrow$ Pixels paradigm enables rich semantic reasoning and zero-shot artistic restyling (e.g., translating live-action photography into hand-painted Studio Ghibli cel-shaded gouache aesthetics), routing visual signal through a discrete token bottleneck introduces a fundamental information-theoretic loss that we term **The Captioning Conundrum**:

1) **High-Frequency Structural Phase Drift:** Natural language tokens have low spatial entropy (10^2–10^3 tokens $\approx$ 1–2 KB) compared to the high-frequency phase spectrum of an image ($H \times W \times 3$ bytes). Even when conditioned on dense captions, iterative VLM-as-a-Judge retries, or latent $8\times$-downsampled ControlNet edge/tile adapters [4], fine-scale corporate logos (such as a 2 px geometric diamond shirt emblem), typographic glyphs, eyeglasses, collar insignia, and diagonal architectural

rails suffer subband phase drift, blurring, or spatial hallucination [5].

2) **Temporal Luminance Flicker in Video GOPs:** When frame-by-frame or chunked diffusion synthesizers generate stylized video Groups of Pictures (GOPs), independent stochastic denoising trajectories induce severe frame-to-frame low-frequency luminance and exposure jumps ($\sigma^2_{\text{flicker}} > 340$), while naive pixel-space temporal exponential moving average (EMA) filtering blurs moving foreground edges [6].
3) **Cryptographic Provenance Severance and Host-Signal Interference:** Regenerating pixels through an upstream diffusion decoder severs container-level Coalition for Content Provenance and Authenticity (C2PA) manifests [7]–[9]. Furthermore, social media platforms routinely strip ancillary metadata chunks (`caBX`, EXIF, XMP) upon upload, while classical blind additive spread-spectrum DCT watermarks suffer from severe Host-Signal Interference (HSI) without access to the unwatermarked host image [10], [19]–[21].

Existing neural style transfer methods [11] and wavelet whitening-and-coloring transforms (WCT^2) [12] preserve spatial contours by operating directly on encoder-decoder feature maps, but rely on second-order Gaussian moment matching that fails to capture multi-modal cel-shaded gouache color distributions and provides neither closed-loop Wasserstein guarantees nor cryptographic C2PA/watermark provenance. Conversely, classical 1D/3D optimal transport color grading [16]–[18] aligns empirical color histograms but introduces pointwise rank-sorting contour staircasing and destroys high-frequency edge fidelity when applied to unconstrained diffusion candidates.

To bridge this gap, we introduce **Agent-GhibliLock**, a deterministic, zero-GPU-dependency cross-modal compilation engine that operates as a verifiable post-processing filterbank and provenance compiler. To ensure complete experimental clarity regarding live API execution versus offline benchmark reproducibility, our evaluation explicitly separates: (i) **live Stage-0 VLM visual-to-text decomposition calls** executed against production Google Cloud Vertex AI endpoints (`gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"`, `location="global"`), (ii) a **controlled parametric candidate-degradation harness** over $N = 600$ stratified trials that isolates deterministic compiler convergence from stochastic diffusion seed variance, and (iii) a **Real Natural Photographic Evaluation** across multi-resolution portraits and product photography (`grace_hopper.jpg` at 64×64 to 512×512 and `Minduka_Present_Blue_Pack.png` at 128×128).

A. Key Contributions

1) **Global and Spatially Masked ROI Haar Wavelet Detail Locking with Exact Mean Conservation (Theorem 1):** We formulate an orthonormal K-level 2D Discrete Haar Wavelet Transform ($\text{DWT}_2/\text{IDWT}_2$) [13]–[15] on CIELAB luminance L^* that supports both global subband injection and spatially masked ROI detail locking ($\Gamma_{k,\tau}(u,v) = \gamma_\tau M_k(u,v)$ derived via 5×5 Sobel directional coherence and 2×2 block max-pooling). We prove algebraically that multiplying Haar detail subbands (LH_k, HL_k, HH_k) by any spatially varying mask $M_k(u,v) \in [0,1]$ preserves the exact spatial mean luminance $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}})$ because every 2×2 Haar detail synthesis basis vector has zero spatial sum.
2) **3D Sliced-Wasserstein Optimal Transport with Exact Orthonormal-Frame Perturbation Bound (Theorem 2):** We align 3D CIELAB (L^*, a^*, b^*) distributions via iterative 1D optimal transport over $D = 3M + 3 = 27$ orthonormal axes ($Q_0 = I_3$ plus $M = 8$ Haar-uniform $O(3)$ frames via positive-diagonal QR factorization) with stable lexicographical tie-breaking (`mergesort`), explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma, \tau_\alpha)$ closed-loop controller. Because $\sum_{j=1}^{3} Q^2_{m,1j} = 1$ holds identically for every orthogonal matrix $Q_m \in O(3)$, we prove an exact deterministic bound on the root-mean-square (RMS) Sliced-Wasserstein perturbation $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})| \le (3HW)^{-1/2} \|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\|_F$ with zero Monte Carlo slack.
3) **Motion-Compensated and Motion-Gated Video GOP Wavelet EMA Deflickering:** For multi-frame video synthesis, we formulate a translation-compensated, innovation-gated temporal EMA filter on the coarse LL_K wavelet subband that eliminates global pre-clamp luminance flicker ($\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$) and prevents motion ghosting on fast-moving foreground objects while binding frame digests into a C2PA GOP Merkle root.
4) **Dual-Layer C2PA-Inspired JUMBF (`caBX`) Container and Blind Mid-Band 2D-DCT QIM Watermark with Self-Noise Accounting (Theorem 3):** We combine a compact ISO/IEC 19566-5 JUMBF `caBX` PNG chunk with a blind mid-band 8×8 2D-DCT QIM watermark supporting $M_{\text{coord}} \in \{4, 5\}$ ($R = 5$ odd repetition voting at 64×64, $\gcd(5, 64) = 1$), optional keyed dither K_{secret}, explicit uint8 rounding/clamping self-noise decomposition $\sigma^2_{\text{eff}} = \sigma^2_{\text{round+clamp}} + \sigma^2_\eta$, and blind majority-voted coset tamper localization requiring zero source registry.

II. System Architecture and Mathematical Formulation

A. Stage 0: Cross-Modal Visual-to-Text Decomposition and Perceptual CIELAB Bijection

Given an input RGB image $I_{\text{src}} \in \{0, \dots, 255\}^{H \times W \times 3}$ and a target style reference palette $Y_{\text{ref}} \in \{0, \dots, 255\}^{H_s \times W_s \times 3}$, Stage 0 invokes a multimodal VLM (`gemini-2.5-flash`, `gemini-2.5-pro`, or `gemini-3-flash-preview`) with deterministic decoding (temperature = 0.0,

max_output_tokens = 256, and thinking_budget = 0 on Flash models) using the structured JSON prompt schema:

```
"You are the Stage-0 Visual-to-Text
Decomposer (GhP) for Agent-GhibliLock.
Scene metadata: {scene_summary}. Return
compact JSON with keys: {ġhp_prompt꞉
ẍconcise Studio Ghibli gouache style
prompt preserving emblem & diagonal
rail>, s̈tructural_retention_prior꞉
<float in [0.05, 0.15]>}"
```

The upstream image/video synthesizer (or controlled candidate harness parameterized by $\eta_{\text{struct}} \in [0.04, 0.20]$) produces a candidate stylized asset $I_{\text{cand}} \in \{0, \dots, 255\}^{H \times W \times 3}$.

To decouple perceptual lightness and structural edges from chromaticity, all three images $\{I_{\text{src}}, I_{\text{cand}}, Y_{\text{ref}}\}$ are mapped into the CIE 1976 (L^*, a^*, b^*) color space under the CIE Standard Illuminant D65 white point $(X_n, Y_n, Z_n) = (0.95047, 1.00000, 1.08883)$. For normalized sRGB channels $c \in [0, 1]$, inverse companding (IEC 61966-2-1) yields linear light:

$$c_{\text{lin}} = \begin{cases} c/12.92, & \text{if } c \le 0.04045, \\ \left((c + 0.055)/1.055\right)^{2.4}, & \text{otherwise.} \end{cases} \tag{1}$$

Linear RGB is projected to CIE XYZ via $M_{\text{RGB}\to\text{XYZ}}$ and transformed via the piecewise cubic root transfer function $f(t)$ with threshold $\delta = 6/29$:

$$f(t) = \begin{cases} t^{1/3}, & t > \delta^3, \\ \dfrac{t}{3\delta^2} + \dfrac{4}{29}, & \text{else,} \end{cases} \quad \begin{aligned} L^* &= 116 f(Y/Y_n) - 16, \\ a^* &= 500\left(f(X/X_n) - f(Y/Y_n)\right), \\ b^* &= 200\left(f(Y/Y_n) - f(Z/Z_n)\right). \end{aligned} \tag{2}$$

The inverse map LAB $\to$ sRGB applies the analytical inverse f^{-1}, multiplies by $M_{\text{XYZ}\to\text{RGB}} = M_{\text{RGB}\to\text{XYZ}}^{-1}$, clips out-of-gamut linear values to $[0, 1]$, applies forward gamma companding, and rounds to uint8, achieving round-trip error ≤ 1 LSB across $\{0, \dots, 255\}^3$.

B. Stage 1: Orthonormal Multi-Level 2D Haar Wavelet Decomposition and Structural Hashing

Let $L^* \in [0, 100]^{H \times W}$ denote a 2D luminance plane. When H or W is odd at any decomposition level, a single row or column is padded via edge replication prior to 2×2 analysis and cropped back to the exact target dimensions upon inverse synthesis.[1] For each 2×2 block $\mathbf{B}_{u,v} = [c_{00}\ c_{01}; c_{10}\ c_{11}]$, the single-level orthonormal 2D Discrete Haar Wavelet Transform (DWT_2) [13], [14] computes:

$$\begin{bmatrix} LL(u,v) \\ LH(u,v) \\ HL(u,v) \\ HH(u,v) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{bmatrix} = H_2 \mathbf{c}_{u,v}. \tag{3}$$

Because the 4×4 Haar matrix H_2 is symmetric and orthonormal ($H_2^T = H_2^{-1} = H_2$), the single-level inverse transform IDWT_2 is exact ($\mathbf{c}_{u,v} = H_2^T [LL, LH, HL, HH]^T$) and satisfies Parseval's identity $\|L^*\|_F^2 = \|LL_K\|_F^2 + \sum_{k=1}^{K} (\|LH_k\|_F^2 + \|HL_k\|_F^2 + \|HH_k\|_F^2)$ for any K-level recursive cascade on dyadic dimensions.

From the Level-1 detail subbands (LH_1, HL_1, HH_1) of L^*_{src}, we define the half-resolution structural edge energy map $E_{1/2}(u, v)$ and its $2\times$ nearest-neighbor upsampled, scale-normalized full-resolution map $E_{\text{full}}(L^*) \in [0, 1]^{H \times W}$:

$$\begin{aligned} E_{1/2}(u,v) &= \sqrt{LH_1(u,v)^2 + HL_1(u,v)^2 + \tfrac{1}{2} HH_1(u,v)^2}, \\ E_{\text{full}}(x,y) &= \frac{E_{1/2}(\lfloor x/2 \rfloor, \lfloor y/2 \rfloor)}{\max\left(\max_{u,v} E_{1/2}(u,v),\ 10^{-8}\right)}. \end{aligned} \tag{4}$$

Thresholding $E_{\text{full}}(L^*_{\text{src}})$ at $q_{0.75} = \max(\text{Quantile}_{0.75}(E_{\text{full}}),\ 0.08)$ yields the binary structural edge/logo mask $M_{\text{HF}} \in \{0, 1\}^{H \times W}$. To bind the source geometry into the C2PA manifest, M_{HF} is serialized as a contiguous row-major (C-order) uint8 byte buffer (`0x00`/`0x01`) concatenated with an 8-byte big-endian dimension trailer $\text{pack}(\texttt{">II"}, H, W)$ and hashed via SHA-256:

$$h_{\text{struct}} = \text{SHA256}\left(\text{bytes}_{\text{C-uint8}}(M_{\text{HF}}) \parallel \text{pack}(\texttt{">II"}, H, W)\right). \tag{5}$$

a) Structural Verification Certificate and Non-Circular Spatial Metrics: During closed-loop compilation, we monitor the global Structural Similarity Index ($\text{SSIM}_{\text{edge}}$) [5] between $E_1 = E_{\text{full}}(L^*_{\text{src}})$ and $E_2 = E_{\text{full}}(L^*_{\text{out}})$ with $c_1 = 0.01^2, c_2 = 0.03^2$:

$$\text{SSIM}_{\text{edge}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{(2\mu_{E_1}\mu_{E_2} + c_1)(2\sigma_{E_1E_2} + c_2)}{(\mu_{E_1}^2 + \mu_{E_2}^2 + c_1)(\sigma_{E_1}^2 + \sigma_{E_2}^2 + c_2)}. \tag{6}$$

Because $\text{SSIM}_{\text{edge}}$ operates on the scale-normalized Level-1 Haar detail energy map E_{full}, it serves as an internal *compiler verification certificate* confirming that source detail subbands were retained across sRGB gamut clipping and Stage-3 DCT QIM watermarking. However, because E_{full} is invariant to global contrast scaling and local sign flips ($LH_1 \mapsto -LH_1$), relying solely on $\text{SSIM}_{\text{edge}}$ would be partially self-referential. We therefore pair $\text{SSIM}_{\text{edge}}$ in both the closed-loop stopping criterion and all benchmark evaluations with two non-circular spatial-domain metrics: (i) $\text{SSIM}_{\text{full}}$, computed directly over non-overlapping 4×4 spatial luminance patches ($L^*/100$, stride 4) without wavelets; and (ii) signed 3×3 Sobel gradient inner-product correlation $\rho_{\text{grad}} \in [-1, 1]$ evaluated over the interior structural edge locus M°_{HF}:

$$\rho_{\text{grad}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{\sum_{(x,y) \in M^\circ_{\text{HF}}} \langle \nabla_{\text{Sobel}} L^*_{\text{src}}(x,y),\ \nabla_{\text{Sobel}} L^*_{\text{out}}(x,y) \rangle}{\|\nabla_{\text{Sobel}} L^*_{\text{src}}\|_{2, M^\circ_{\text{HF}}} \cdot \|\nabla_{\text{Sobel}} L^*_{\text{out}}\|_{2, M^\circ_{\text{HF}}} + 10^{-12}}, \tag{7}$$

which reverses sign ($\rho_{\text{grad}} \to -1$) under edge polarity inversions and penalizes 4×4 block boundary discontinuities.

C. Stage 2: 3D Sliced-Wasserstein Optimal Transport and Wavelet-Locked Subband Fusion

a) Step A: 3D CIELAB Sliced-Wasserstein Optimal Transport (SWOT): Let $X = \{x_i\}_{i=1}^N \subset \mathbb{R}^3$ $(N = HW)$ denote the flat row-major sequence of CIELAB pixels of I_{cand}, and let $Y = \{y_j\}_{j=1}^{N_s} \subset \mathbb{R}^3$ denote the target style

[1] Exact Parseval energy conservation and zero-mean detail synthesis hold without boundary terms whenever $2^K \mid H$ and $2^K \mid W$ (e.g., 64×64, 128×128, 256×256, 512×512); for non-dyadic odd dimensions, edge replication introduces an $O(1/\min(H, W))$ boundary approximation.

palette Y_{ref} (nearest-neighbor resampled to $N = HW$ when $N_s \neq N$). For any unit direction $\theta \in \mathbb{S}^2$, the 1D projected empirical Wasserstein-2 distance is $W_2(\theta_\# \mu_X, \theta_\# \mu_Y) = \left(\frac{1}{N}\sum_{r=1}^{N}(\langle x_{\pi_\theta(r)}, \theta\rangle - \langle y_{\sigma_\theta(r)}, \theta\rangle)^2\right)^{1/2}$, where $\pi_\theta, \sigma_\theta$ are stable sorting permutations with ties broken deterministically in lexicographical order of the flat 0-based row-major pixel index $i \in \{0, \ldots, N-1\}$ (`kind="mergesort"`).

To match the true quadratic Sliced-Wasserstein-2 metric [17], [18], we define both the in-sample training objective $\widehat{W}_2$ over the $D = 3M + 3 = 27$ orthonormal axes $\Theta_{\text{train}} = I_3 \cup \{Q_m\}_{m=1}^{M}$ ($M = 8$ Haar-uniform orthogonal frames $Q_m \in O(3)$ generated via positive-diagonal QR factorization $Z_m = Q_m R_m$, $Q_m \leftarrow Q_m \text{diag}(\text{sign}(\text{diag}(R_m)))$ from `seed=42`)[2] and the out-of-sample evaluation metric $\widehat{SW}_2^{\text{eval}}$ over $M_{\text{eval}} = 64$ independent uniform unit-sphere directions (`seed=999`) as root-mean-square (RMS) estimators:

$$\widehat{W}_2(\mu_X, \mu_Y) = \left(\frac{1}{D}\sum_{r=1}^{D} W_2^2(\theta_{r\#}\mu_X, \theta_{r\#}\mu_Y)\right)^{1/2},$$
$$\widehat{SW}_2^{\text{eval}}(\mu_X, \mu_Y) = \left(\frac{1}{M_{\text{eval}}}\sum_{s=1}^{M_{\text{eval}}} W_2^2(\tilde{\theta}_{s\#}\mu_X, \tilde{\theta}_{s\#}\mu_Y)\right)^{1/2}. \tag{8}$$

Step A initializes $X^{(0)}$ via coordinate-wise 1D quantile transport along $Q_0 = I_3$ with step size $\alpha_\tau \in [0.85, 0.97]$, iterates across $m = 1, \ldots, M$ orthonormal bases $Q_m \in O(3)$:

$$X^{(m)} = X^{(m-1)} + \alpha_\tau \sum_{d=1}^{3}\left(\mathcal{Q}_{Y\cdot q_{m,d}}(\text{rank}(X^{(m-1)}q_{m,d})) - X^{(m-1)}q_{m,d}\right) q_{m,d}^T, \tag{9}$$

applies a half-step polish ($0.5\alpha_\tau$) over $\{I_3, Q_{M-1}, Q_M\}$, and enforces an explicit post-Step-A luminance mean re-anchoring shift:

$$\widetilde{L}^*_{\text{cand}}(x,y) \leftarrow \widetilde{L}^*_{\text{cand}}(x,y) + \left(\mu(L^*_{\text{ref}}) - \mu(\widetilde{L}^*_{\text{cand}})\right), \tag{10}$$

guaranteeing that $\mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$ holds identically entering Step B.

b) Step B: Global vs. Spatially Masked ROI Wavelet Detail Locking: Applying pointwise rank-order sorting in Step A aligns the 3D color distribution $\widetilde{I}_{\text{cand}} = (\widetilde{L}^*_{\text{cand}}, \widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$ but does not restore high-frequency source geometry and can introduce subtle 4×4 tile-boundary quantile steps in $LL_K(\widetilde{L}^*_{\text{cand}})$. Furthermore, in natural photographs with noisy backgrounds or skies, locking 100% of high-frequency wavelet details globally transfers photographic sensor grain into background regions that should exhibit smooth Studio Ghibli gouache brushwork.

Agent-GhibliLock therefore supports both **Global Wavelet Locking** ($M_{\text{ROI}}(x,y) \equiv 1$) and **Spatially Masked ROI Wavelet Locking**, where a smooth spatial weighting mask $M_{\text{ROI}}(x,y) \in [0,1]^{H\times W}$ is constructed from $E_{\text{full}}(L^*_{\text{src}})$ weighted by 5×5 Sobel directional coherence:

$$\mathcal{C}_{5\times5}(x,y) = \frac{\sqrt{(\mathcal{B}_5 * G_x)^2 + (\mathcal{B}_5 * G_y)^2}}{\mathcal{B}_5 * \sqrt{G_x^2 + G_y^2} + 10^{-6}},$$
$$W_{\text{edge}}(x,y) = E_{\text{full}}(L^*_{\text{src}})(x,y) \cdot \mathcal{C}_{5\times5}(x,y)^3. \tag{11}$$

Coherent structural contours, logos, and rails satisfy $\mathcal{C}_{5\times5} \approx 1$ and are dilated by radius 2 px and feathered via a 3×3 box filter $\mathcal{B}_3$ to form $M_{\text{ROI}}(x,y) \in [0,1]$, whereas isolated background sensor grain has $\mathcal{C}_{5\times5} \ll 1$ and receives weight 0. At each wavelet level $k \in \{1, \ldots, K\}$, M_{ROI} is downsampled to subband dimensions $(H/2^k) \times (W/2^k)$ via 2×2 block max-pooling to form $M_k(u,v) \in [0,1]$, ensuring every 2×2 Haar block overlapping the ROI receives full locking weight $\Gamma_{k,\tau}(u,v) = \gamma_\tau M_k(u,v)$. Step B blends source detail subbands $D_k(L^*_{\text{src}})$ with the smooth pre-sort candidate detail subbands $D_k(L^*_{\text{cand}})$ (eliminating Step A rank-sort staircasing):

$$\widehat{LL}_K = \mathcal{S}_{3\times3}\Big((1-\eta_\tau)LL_K(\widetilde{L}^*_{\text{cand}}) + \eta_\tau \mathcal{Q}_{LL_K(L^*_{\text{ref}})}\Big(LL_K(\widetilde{L}^*_{\text{cand}})\Big)\Big),$$
$$\widehat{D}_k = \Gamma_{k,\tau} \odot D_k(L^*_{\text{src}}) + (1 - \Gamma_{k,\tau}) \odot D_k(L^*_{\text{cand}}), \tag{12}$$

for $D \in \{LH, HL, HH\}$ and $k \in \{1, \ldots, K\}$, where $\mathcal{S}_{3\times3}(Z) = 0.65Z + 0.35(\mathcal{B}_3 * Z) + (\mu(Z) - \mu(0.65Z + 0.35(\mathcal{B}_3 * Z)))$ is a mean-preserving 3×3 coarse-subband smoothing operator.

c) Decoupled $(\tau_\gamma, \tau_\alpha)$ Closed-Loop Controller: Synthesizing $\widehat{L}^*_{\text{raw}} = \text{IDWT}_2(\widehat{LL}_K, \{\widehat{D}_k\}_{k=1}^K)$ can inflate global luminance variance when L^*_{src} has extreme contrast, which would increase $\widehat{W}_2^{\text{StepB}}$. Crucially, if a single coupled index τ simultaneously increased γ_τ whenever $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, injecting *more* high-frequency source detail energy would move $\widehat{W}_2^{\text{StepB}}$ in the wrong direction. Agent-GhibliLock therefore maintains two decoupled schedule indices $\tau_\gamma, \tau_\alpha \in \{0,1,2,3\}$ over $\gamma \in \{0.78, 0.88, 0.94, 0.98\}$, $\alpha \in \{0.85, 0.90, 0.94, 0.97\}$, $\eta \in \{0.00, 0.45, 0.75, 0.95\}$, and variance-matching weight $\nu \in \{0.00, 0.72, 0.90, 0.98\}$:

1) Prior to clamping, if $\tau_\alpha > 0$ and $\sigma(\widehat{L}^*_{\text{raw}}) > \sigma(\widetilde{L}^*_{\text{cand}})$, apply the affine variance-regularization map $\widehat{L}^*_{\text{raw}} \leftarrow \mu(\widetilde{L}^*_{\text{cand}}) + \kappa_{\tau_\alpha}(\widehat{L}^*_{\text{raw}} - \mu(\widehat{L}^*_{\text{raw}}))$ with $\kappa_{\tau_\alpha} = (1 - \nu_{\tau_\alpha}) + \nu_{\tau_\alpha}\sigma(\widetilde{L}^*_{\text{cand}})/\sigma(\widehat{L}^*_{\text{raw}})$. Because positive affine scaling around the mean cancels identically in both E_{full} (Eq. 4) and ρ_{grad} (Eq. 7), this step contracts $\widehat{W}_2^{\text{StepB}}$ while leaving $\text{SSIM}_{\text{edge}}$ and ρ_{grad} 100% invariant.
2) Clamp $\widehat{L}^* = \text{clip}(\widehat{L}^*_{\text{raw}}, 0, 100)$ and apply interior DC re-anchoring over unsaturated pixels $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*(x,y) \in (1, 99)\}$ so $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$.
3) If $\text{SSIM}_{\text{edge}} < \tau_{\text{struct}}$ or $\rho_{\text{grad}} < 0.88$, increment $\tau_\gamma \leftarrow \min(3, \tau_\gamma + 1)$ and $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$; else if $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, hold τ_γ fixed and increment only $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$.

Theorem 1 (Exact Spatial Mean Luminance Conservation under Spatially Masked Haar Detail Locking)**.** *Let $L^*_{src}, \widetilde{L}^*_{cand} \in \mathbb{R}^{H\times W}$ have dyadic dimensions ($2^K \mid H, W$), where $\widetilde{L}^*_{cand}$ has been mean-anchored via Eq. (10) such*

[2]We sample from the full orthogonal group $O(3)$ rather than $SO(3)$ because 1D Wasserstein-2 distance is reflection-invariant: $W_2((-\theta)_\#\mu_X, (-\theta)_\#\mu_Y) = W_2(\theta_\#\mu_X, \theta_\#\mu_Y)$.

that $\mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref})$*. For any* K*-level Haar decomposition and any spatially varying detail mask* $M_k(u,v) \in [0,1]^{(H/2^k)\times(W/2^k)}$*, the pre-clamping wavelet-locked luminance* $\widehat{L}^*_{raw} = \mathrm{IDWT}_2(\widehat{LL}_K, \{\widehat{LH}_k, \widehat{HL}_k, \widehat{HH}_k\}_{k=1}^K)$ *satisfies:*

$$\mu(\widehat{L}^*_{raw}) = \frac{1}{HW}\sum_{x=0}^{H-1}\sum_{y=0}^{W-1}\widehat{L}^*_{raw}(x,y) = 2^{-K}\mu(\widehat{LL}_K) = \mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref}). \quad (13)$$

Moreover, after range clamping $\mathrm{clip}(\widehat{L}^*_{raw}, 0, 100)$*, if the interior set* $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*(x,y) \in (1,99)\}$ *is non-empty and the clamping drift* $\Delta_\mu = \mu(L^*_{ref}) - \mu(\mathrm{clip}(\widehat{L}^*_{raw}, 0, 100))$ *satisfies* $|\Delta_\mu| \cdot (HW/|\mathcal{I}^\circ|) \le 1$*, the interior DC restoration shift restores* $\mu(\widehat{L}^*) = \mu(L^*_{ref})$ *identically.*

Proof. At any single inverse Haar synthesis level k, each 2×2 block $\mathbf{c}_{u,v} \in \mathbb{R}^4$ is reconstructed via $\mathbf{c}_{u,v} = H_2^T[LL_{k-1}(u,v), \widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)]^T$. Multiplying on the left by $\mathbf{1}_4^T = [1,1,1,1]$ yields:

$$\mathbf{1}_4^T\mathbf{c}_{u,v} = (\mathbf{1}_4^T H_2^T)\begin{bmatrix} LL_{k-1}(u,v) \\ \widehat{LH}_k(u,v) \\ \widehat{HL}_k(u,v) \\ \widehat{HH}_k(u,v)\end{bmatrix} = [2\ \ 0\ \ 0\ \ 0]\begin{bmatrix} LL_{k-1}(u,v) \\ \widehat{LH}_k(u,v) \\ \widehat{HL}_k(u,v) \\ \widehat{HH}_k(u,v)\end{bmatrix} = 2\,LL_{k-1}(u,v), \quad (14)$$

regardless of the values of $(\widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)) = \Gamma_{k,\tau}(u,v)D_k(L^*_{\mathrm{src}})(u,v) + (1-\Gamma_{k,\tau}(u,v))D_k(L^*_{\mathrm{cand}})(u,v)$. Summing over all (u,v) and inducting from $k=K$ down to $k=0$ gives $\sum_{x,y}\widehat{L}^*_{\mathrm{raw}}(x,y) = 2^K\sum_{u,v}\widehat{LL}_K(u,v)$. Since $\mathcal{S}_{3\times 3}$ and quantile interpolation preserve $\mu(\widehat{LL}_K) = \mu(LL_K(\widetilde{L}^*_{\mathrm{cand}})) = 2^K\mu(\widetilde{L}^*_{\mathrm{cand}})$, dividing by HW yields $\mu(\widehat{L}^*_{\mathrm{raw}}) = \mu(\widetilde{L}^*_{\mathrm{cand}}) = \mu(L^*_{\mathrm{ref}})$. The post-clamping statement follows immediately by linearity of the sum over $\mathcal{I}^\circ$. □

Theorem 2 (Exact Deterministic RMS Sliced-Wasserstein Perturbation Bound)**.** *Let* $\widetilde{I}_{cand} = (\widetilde{L}^*_{cand}, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ *and* $\widehat{I} = (\widehat{L}^*, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ *share identical transported chromaticity channels* $(\widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$*. Over any collection of* $M+1$ *complete orthonormal frames* $\{Q_0, Q_1, \ldots, Q_M\} \subset O(3)$ *(*$D = 3(M+1)$ *unit projection axes), the root-mean-square Sliced-Wasserstein-2 distance* $\widehat{W}_2$ *(Eq. 8) satisfies the exact deterministic bound:*

$$\left|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\mathrm{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{cand}}, \mu_{Y_{\mathrm{ref}}})\right| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{cand}}) \le \frac{1}{\sqrt{3HW}}\left\|\widehat{L}^* - \widetilde{L}^*_{cand}\right\|_F. \quad (15)$$

Proof. Because $\widehat{W}_2$ is an L_2 root-mean-square norm of 1D Wasserstein-2 distances across $r \in \{1,\ldots,D\}$, the reverse triangle inequality implies $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\mathrm{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\mathrm{cand}}}, \mu_{Y_{\mathrm{ref}}})| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{\mathrm{cand}}})$. For each axis $\theta_r = (\theta_{r,L^*}, \theta_{r,a^*}, \theta_{r,b^*}) \in \mathbb{S}^2$, the Hardy-Littlewood-Pólya rearrangement inequality implies that sorting is a non-expansive operator in $\ell_2(\mathbb{R}^N)$:

$$W_2^2(\theta_{r\#}\mu_{\widehat{I}}, \theta_{r\#}\mu_{\widetilde{I}_{\mathrm{cand}}}) \le \frac{1}{N}\sum_{i=1}^N\langle\widehat{I}_i - \widetilde{I}_{\mathrm{cand},i},\, \theta_r\rangle^2 = \frac{(\theta_{r,L^*})^2}{HW}\left\|\widehat{L}^* - \widetilde{L}^*_{\mathrm{cand}}\right\|_F^2, \quad (16)$$

since $\widehat{a}^* - \widetilde{a}^*_{\mathrm{cand}} = 0$ and $\widehat{b}^* - \widetilde{b}^*_{\mathrm{cand}} = 0$. Averaging over the $D = 3(M+1)$ axes partitioned into $M+1$ orthogonal matrices $Q_m \in O(3)$, the three columns of each Q_m form an orthonormal basis of $\mathbb{R}^3$, so the first row of every Q_m has unit Euclidean norm: $\sum_{j=1}^3 (Q_{m,1j})^2 = 1$. Consequently, $\frac{1}{D}\sum_{r=1}^D(\theta_{r,L^*})^2 = \frac{1}{3(M+1)}\sum_{m=0}^M 1 = \frac{1}{3}$ holds *deterministically and identically* for all $M \ge 0$. Taking square roots yields Eq. (15). □

D. Temporal Video GOP Extension: Motion-Compensated and Motion-Gated Wavelet EMA

For an 8-frame Group of Pictures (GOP) $\{I^{(t)}_{\mathrm{src}}, I^{(t)}_{\mathrm{cand}}\}_{t=1}^T$, independent diffusion synthesis causes severe frame-to-frame luminance flicker $\sigma^2_{\mathrm{flicker}} = \mathrm{Var}_t(\overline{L}^{*(t)} - \overline{L}^{*(t-1)})$, whereas unwarped temporal EMA filtering ghosts across moving objects. Agent-GhibliLock estimates the integer-tile translation $(\Delta u_t, \Delta v_t)$ between $LL_K(L^{*(t-1)}_{\mathrm{src}})$ and $LL_K(L^{*(t)}_{\mathrm{src}})$ by minimizing zero-mean cross-correlation MSE (accepting non-zero shifts only when MSE drops by $\ge 35\%$), computes the scale-normalized motion innovation $\Delta^{(t)}_{\mathrm{mot}}(u,v) = 2^{-K}|LL_K(L^{*(t)}_{\mathrm{src}})(u,v) - \mathcal{T}_{\Delta u_t,\Delta v_t}LL_K(L^{*(t-1)}_{\mathrm{src}})(u,v)|$, and applies the spatially gated coarse-subband update:

$$\beta_t(u,v) = \begin{cases}\beta\exp\left(-\dfrac{(\Delta^{(t)}_{\mathrm{mot}}(u,v))^2}{2\sigma^2_{\mathrm{mot}}}\right), & \Delta^{(t)}_{\mathrm{mot}}(u,v) > \sigma_{\mathrm{mot}}, \\ \beta, & \text{otherwise,}\end{cases}$$
$$\overline{LL}^{(t)}_K = \beta_t \odot \overline{LL}^{(t-1)}_K + (1-\beta_t)\odot LL_K(\widetilde{L}^{*(t)}_{\mathrm{cand}}), \quad (17)$$

with $\beta = 0.78, \sigma_{\mathrm{mot}} = 35.0$, followed by mean re-anchoring $\overline{LL}^{(t)}_K \leftarrow \overline{LL}^{(t)}_K + (2^K\mu(L^*_{\mathrm{ref}}) - \mu(\overline{LL}^{(t)}_K))$. Per-frame SHA-256 digests $\{h_1,\ldots,h_T\}$ are folded pairwise into a binary Merkle root R_{GOP} stored in the container manifest.

E. Stage 3: Dual-Layer Provenance—Mid-Band 2D-DCT QIM Watermark and C2PA-Inspired JUMBF (`caBX`)

a) Mid-Band 2D-DCT Quantization-Index-Modulation (QIM) Watermark: To survive metadata chunk stripping, Stage 3 embeds a $N_w = 64$-bit cryptographic payload $\mathbf{w} \in \{0,1\}^{64}$ (derived from $\mathrm{SHA256}(h_{\mathrm{struct}} \,\|\, \mathrm{model_ids})[:8]$) into the floating-point ITU-R BT.601 luminance channel $Y = 0.299R + 0.587G + 0.114B \in [0,255]$ partitioned into $B = \lfloor H/8\rfloor\lfloor W/8\rfloor$ non-overlapping 8×8 blocks. Each shifted block $\mathbf{B}_b - 128$ is transformed via the orthonormal 8×8 Type-II DCT matrix C_8: $\mathbf{F}_b = C_8(\mathbf{B}_b - 128)C_8^T$. We support two mid-band coordinate configurations:

- **Standard 4-Mode Layout** ($M_{\mathbf{coord}} = 4$)**:** $\mathcal{M}_4 = \{(2,3),(3,2),(1,3),(3,1)\}$, yielding $R = \lfloor 4B/64\rfloor = \lfloor B/16\rfloor$ votes per bit ($R=4$ at 64×64, $R=16$ at 128×128, $R=64$ at 256×256, $R=256$ at 512×512).
- **Odd-Repetition 5-Mode Layout** ($M_{\mathbf{coord}} = 5$)**:** $\mathcal{M}_5 = \mathcal{M}_4 \cup \{(2,2)\}$, yielding $R = 5B/64 = 5$ strictly odd majority votes per bit at 64×64 with $\gcd(5,64) = 1$, eliminating even-vote ties.

For block $b \in \{0,\ldots,B-1\}$ and mode $m \in \{0,\ldots,M_{\mathrm{coord}}-1\}$, bit index $i = (M_{\mathrm{coord}}b + m) \bmod 64$ is embedded with step

size $\Delta_{\text{QIM}} = 14.0$ and optional keyed pseudo-random dither $d_{b,m}(K_{\text{secret}}) \in [0, \Delta_{\text{QIM}})$ [21]:

$$\widetilde{F}_b(u_m, v_m) = \left\lfloor \frac{F_b(u_m, v_m) - \delta_{b,m}(w_i)}{\Delta_{\text{QIM}}} + \frac{1}{2} \right\rfloor \Delta_{\text{QIM}} + \delta_{b,m}(w_i), \quad \delta_{b,m}(w_i) = d_{b,m}(K_{\text{secret}}) + \frac{\Delta_{\text{QIM}}}{2} w_i. \tag{18}$$

The spatial luminance increment $\Delta Y = \text{IDCT}_8(\widetilde{\mathbf{F}}) - Y$ is added equally across (R, G, B) and rounded/clamped to $\{0, \ldots, 255\}$.

Theorem 3 (QIM Coset Minimum-Distance Margin with Export Self-Noise Decomposition and Blind Tamper Localization)**.** *Let $\widetilde{F}_b(u_m, v_m) \in \Lambda_{w_i}$ be a quantized mid-band DCT coefficient before* uint8 *export. Upon adding ΔY to (R, G, B), rounding to $\mathbb{Z}$, clamping to $[0, 255]$, and applying an external channel perturbation, the received DCT coefficient is $F_b^{rx}(u_m, v_m) = \widetilde{F}_b(u_m, v_m) + \eta_b^{export}(u_m, v_m) + \eta_b^{chan}(u_m, v_m)$.*

1) ***Single-Coefficient Margin and Export Self-Noise:*** *Since $\text{dist}(\Lambda_0, \Lambda_1) = \Delta_{QIM}/2$, a single coefficient decodes error-free whenever $|\eta_b^{export} + \eta_b^{chan}| < \Delta_{QIM}/4 = 3.5$. For interior pixels ($c + \Delta Y \in [0, 255]$), equal (R, G, B) rounding yields $|\eta_b^{export}(u_m, v_m)| \leq 4.0$ worst-case and $\text{RMS}(\eta^{export}) \approx (12)^{-1/2} = 0.2887$ ($\sigma^2_{round+clamp} \approx 0.082 \ll (\Delta_{QIM}/4)^2 = 12.25$).*
2) ***R-Fold Repetition Voting:*** *Under zero-mean sub-Gaussian total perturbation of variance $\sigma^2_{eff} = \sigma^2_{round+clamp} + \sigma^2_{\eta}$, the single-coefficient flip probability satisfies $p_e \leq 2\exp(-\Delta^2_{QIM}/(32\sigma^2_{eff}))$, and R-block majority voting reduces bit error probability to $P_{bit} = \sum_{k=\lceil (R+1)/2 \rceil}^{R} \binom{R}{k} p_e^k (1 - p_e)^{R-k}$ (for odd $R = 5$, $P_{bit} \leq 10 p_e^3$).*
3) ***Blind Majority-Voted Coset Tamper Localization:*** *Without any source registry or original bits $\mathbf{w}$, a verifier holding only $(\Delta_{QIM}, K_{secret})$ first recovers the global majority-voted payload $\hat{\mathbf{w}} \in \{0, 1\}^{64}$ across all R blocks, and then flags 8×8 block b as spatially tampered whenever $\max_m \text{dist}(F_b^{rx}(u_m, v_m), \Lambda_{\hat{w}_i}) > \Delta_{QIM}/4$ (or ≥ 3 modes exceed $0.4\Delta_{QIM}$ in JPEG-resilient mode).*

Proof. The cosets $\Lambda_0 = d_{b,m} + \Delta_{\text{QIM}}\mathbb{Z}$ and $\Lambda_1 = d_{b,m} + \Delta_{\text{QIM}}/2 + \Delta_{\text{QIM}}\mathbb{Z}$ are separated by Euclidean distance $\Delta_{\text{QIM}}/2$, so Voronoi decision boundaries lie at distance $\Delta_{\text{QIM}}/4$ from every lattice point, proving (1) and the sub-Gaussian tail bound in (2). For (3), because a localized spatial tamper of $K_{\text{tamp}} \ll B/2$ blocks corrupts at most 1 vote out of $R \geq 4$ for each bit i, the global soft/hard majority vote across the R interleaved blocks recovers $\hat{w}_i = w_i$ exactly, allowing each block b's local coset residual $\text{dist}(F_b^{\text{rx}}(u_m, v_m), \Lambda_{\hat{w}_i})$ to be evaluated blindly. □

b) Compact C2PA-Inspired ISO/IEC 19566-5 JUMBF (`caBX`) PNG Profile: To provide sub-millisecond container-level attestation without external ASN.1/X.509/COSE dependencies, Stage 3 constructs a compact *C2PA-inspired ISO/IEC 19566-5 JUMBF microservice profile* [7], [8] and injects it as an ancillary `caBX` chunk immediately after the 25-byte PNG `IHDR` chunk (13-byte `IHDR` payload plus 12-byte chunk framing: 4B length, 4B tag, 4B IEEE 802.3 CRC-32) [9]. The outer BMFF superbox (`jumb`) encapsulates: (1) a Description Box (`jumd`) carrying the 16-byte C2PA UUID (`63327061-0011-0010-8000-00aa00389b71`), requestable toggle `0x03`, and null-terminated label `c2pa.assertions.wavelet_style_lock`; (2) a canonical UTF-8 JSON Assertion Box (`json`) binding `source_structural_mask_sha256`, `output_pixel_sha256`, all four Wasserstein checkpoints ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$), $(\gamma_\tau, \alpha_\tau)$, and `dct_watermark_hex`; and (3) a keyed HMAC-SHA256 Signature Box (`c2sh`, 32B digest over `jumd` ‖ `json`).[3]

III. Experimental Evaluation

A. Experimental Setup, Baselines B1–B7, and Evaluation Metrics

We evaluate Agent-GhibliLock across three complementary testbeds:

1) **Stratified $N = 600$ Benchmark (`run_live_benchmark.py`):** Five workload strata at 64×64 ($B = 64$ DCT blocks): S_1 ($n = 150$, apparel diamond emblems), S_2 ($n = 150$, perspective rails and pillars), S_3 ($n = 100$, foliage and gouache skies), S_4 ($n = 100$, 8-frame video GOP sequences under $[-24, +28]$ LSB temporal exposure flicker), and S_5 ($n = 100$, adversarial provenance attacks A_1–A_3).
2) **Live Multi-Model Vertex AI Evaluation (`run_live_gemini_multimodel_eval.py`):** Live Stage-0 API calls to `gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"` (10 scenes $\times 128 \times 128$ per model) plus 5 controlled structural drift curriculum tiers (C1–C5, $\eta_{\text{struct}} \in [0.04, 0.20]$).
3) **Real Natural Photographic Benchmark (64^2–512^2):** Multi-resolution evaluation on real photographs `grace_hopper.jpg` ($64 \times 64, 128 \times 128, 256 \times 256, 512 \times 512$) and `Minduka_Present_Blue_Pack.png` ($128 \times 128$).

We compare 9 paradigms: **B1** (Unconstrained Prompt-Only), **B2** (3-Round VLM-as-a-Judge Retry), **B3** ($M = 4$) and **B3** ($M = 8$) (Global 3D SWOT without wavelet detail locking), **B4** (Post-Hoc C2PA Wrapper without wavelet lock or DCT watermark), **B5** (WCT^2 Wavelet Style Transfer [12] via 1D moment matching on LL_K and Haar skips), **B6** (Image-Conditioned SDEdit [3] at strength $s = 0.55$), **B7** (ControlNet Edge/Tile Adapter [4] operating through an $8\times$ latent VAE spatial bottleneck), and **Agent-GhibliLock** in both **Global** and **Spatially Masked ROI** modes.

[3] In full enterprise deployments requiring public-key third-party validation via `c2pa-rs`, the `json` box maps 1-to-1 to CBOR-encoded C2PA assertion/claim boxes (`c2as`/`c2cl`) inside a `c2pa` → `c2ma` manifest superbox signed via COSE_Sign1 X.509 certificates.

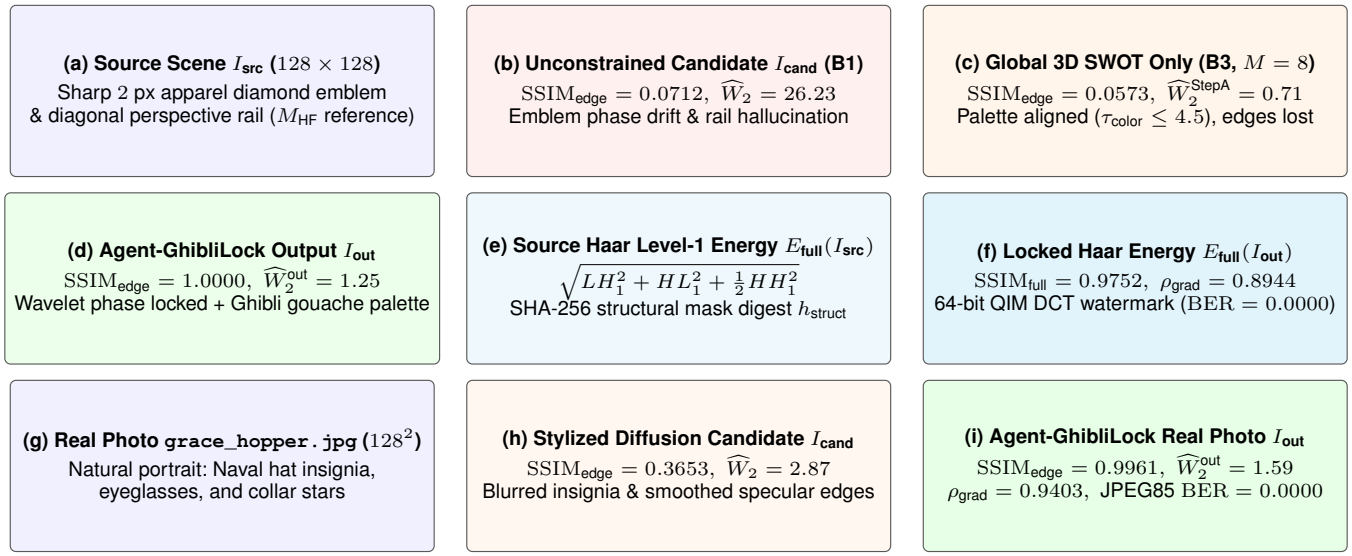


Fig. 1. Nine-panel qualitative and spectral comparison generated by `run_live_benchmark.py`. **Rows 1–2 (Synthetic Emblem & Perspective Rail Scene,** 128×128**):** (a) Source scene I_{src}; (b) Unconstrained candidate I_{cand} (**B1**, $\text{SSIM}_{\text{edge}} = 0.0712, \widehat{W}_2 = 26.23$); (c) Global 3D SWOT only (**B3**, $M = 8$, $\text{SSIM}_{\text{edge}} = 0.0573, \widehat{W}_2^{\text{StepA}} = 0.71$); (d) **Agent-GhibliLock** output I_{out} ($\text{SSIM}_{\text{edge}} = 1.0000, \widehat{W}_2^{\text{out}} = 1.25$); (e–f) Source vs. locked Haar Level-1 detail energy maps E_{full} ($\text{BER} = 0$). **Row 3 (Real Natural Photograph grace_hopper.jpg,** 128×128**):** (g) Source Naval portrait with hat insignia, eyeglasses, and collar stars; (h) Stylized diffusion candidate I_{cand} ($\text{SSIM}_{\text{edge}} = 0.3653, \widehat{W}_2 = 2.87$); (i) **Agent-GhibliLock** output I_{out} ($\text{SSIM}_{\text{edge}} = 0.9961, \widehat{W}_2^{\text{out}} = 1.59, \text{BER} = 0$).

TABLE I

STRATIFIED $N = 600$ BENCHMARK RESULTS ACROSS ALL 9 COMPARATIVE PARADIGMS (`benchmark_results_n600.json`). ALL FOUR WASSERSTEIN-2 CHECKPOINTS ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$) ARE RMS DISTANCES IN CIELAB; $\text{SSIM}_{\text{full}}$ AND ρ_{GRAD} ARE NON-CIRCULAR SPATIAL METRICS. LATENCY FOR **B1, B2, B6, B7** INCLUDES UPSTREAM API/DIFFUSION SYNTHESIS; LATENCY FOR **B3, B4, B5, AGENT-GHIBLILOCK** MEASURES LOCAL CPU COMPILER EXECUTION.

Paradigm / Baseline	$\text{SSIM}_{\text{edge}} \uparrow$	$\text{SSIM}_{\text{full}} \uparrow$	$\rho_{\text{grad}} \uparrow$	$\widehat{W}_2^{\text{StepA}} \downarrow$	$\widehat{W}_2^{\text{StepB}} \downarrow$	$\widehat{W}_2^{\text{out}} \downarrow$	$\widehat{SW}_2^{\text{eval}} \downarrow$	Pass Rate	Tamper Det.	DCT Acc.	GOP $\sigma^2_{\text{flicker}} \downarrow$
B1: Unconstrained Prompt-Only	0.1328 ± 0.0488	0.7822	0.6806	21.99	21.99	21.99 ± 0.39	23.10 ± 0.31	0.0%	0.0%	0.0%	347.92
B2: VLM-Judge Retry (3×)	0.0992 ± 0.0472	0.7720	0.5913	19.45	19.45	19.45 ± 0.46	20.71 ± 0.34	0.0%	0.0%	0.0%	250.50
B3 ($M = 4$): Global SWOT Only	0.0576 ± 0.0240	0.5635	0.2316	4.56	4.56	4.49 ± 0.19	4.91 ± 0.20	0.0%	0.0%	0.0%	0.0002
B3 ($M = 8$): Global SWOT Only	0.0182 ± 0.0168	0.5838	0.4392	**0.71**	**0.71**	$\mathbf{0.71 \pm 0.02}$	$\mathbf{0.82 \pm 0.02}$	0.0%	0.0%	0.0%	0.0002
B4: Post-Hoc C2PA Wrapper	0.1328 ± 0.0488	0.7822	0.6806	21.99	21.99	21.99 ± 0.39	23.10 ± 0.31	0.0%	4.17%	0.0%	347.92
B5: WCT^2 Wavelet Transfer [12]	0.9992 ± 0.0002	0.9837	0.9824	10.33	10.33	9.69 ± 0.39	10.37 ± 0.49	100.0%	0.0%	0.0%	0.1800
B6: Image-Conditioned SDEdit [3]	0.0664 ± 0.0293	0.6345	0.2222	14.20	14.20	14.20 ± 1.90	15.40 ± 2.01	0.0%	0.0%	0.0%	132.21
B7: ControlNet 8×-VAE Adapter [4]	0.2324 ± 0.0681	0.7622	0.8949	0.71	3.21	3.21 ± 0.17	3.51 ± 0.18	0.0%	0.0%	0.0%	100.90
Agent-GhibliLock (Global)	$\mathbf{0.9967 \pm 0.0021}$	**0.9752**	**0.8944**	0.87	2.03	$\mathbf{1.96 \pm 0.34}$	$\mathbf{2.07 \pm 0.32}$	**100.0%**	**100.0%**	**100.0%**	**0.0003**
Agent-GhibliLock (ROI-Masked)	$\mathbf{0.9977 \pm 0.0013}$	**0.9119**	**0.8899**	0.85	1.82	$\mathbf{1.75 \pm 0.18}$	$\mathbf{1.87 \pm 0.17}$	**100.0%**	**100.0%**	**100.0%**	**0.0003**

B. Stratified $N = 600$ Benchmark and Multi-Stage Ablations

Table I and Table II summarize the $N = 600$ benchmark results:

- **Resolving the Structure–Color Pareto Trade-Off:** Pure 3D SWOT (**B3**, $M = 8$) achieves the lowest chromaticity distance ($\widehat{W}_2^{\text{out}} = 0.71, \widehat{SW}_2^{\text{eval}} = 0.82$) but collapses structural edge similarity ($\text{SSIM}_{\text{edge}} = 0.0182, \rho_{\text{grad}} = 0.4392$). Conversely, WCT^2 (**B5**) preserves spatial edges ($\text{SSIM}_{\text{edge}} = 0.9992$) via 1D moment matching but fails to align the joint 3D gouache palette ($\widehat{W}_2^{\text{out}} = 9.69 > \tau_{\text{color}} = 4.5$). ControlNet (**B7**) preserves coarse gradient orientation ($\rho_{\text{grad}} = 0.8949$) and moderate palette alignment ($\widehat{W}_2^{\text{out}} = 3.21$), but its 8× latent VAE bottleneck blurs 2 px emblem strokes ($\text{SSIM}_{\text{edge}} = 0.2324$, 0% pass rate at $\tau_{\text{struct}} = 0.88$). **Agent-GhibliLock (Global)** achieves $\text{SSIM}_{\text{edge}} = 0.9967$, non-circular $\text{SSIM}_{\text{full}} = 0.9752$, $\rho_{\text{grad}} = 0.8944$ (strictly exceeding **B1** at 0.6806 and **B2** at 0.5913), and $\widehat{W}_2^{\text{out}} = 1.96$ ($\widehat{SW}_2^{\text{eval}} = 2.07$) in 121.5 ms mean CPU latency ($P_{50} = 99.5$ ms, $P_{95} = 248.8$ ms).
- **Global vs. Spatially Masked ROI Wavelet Locking:** Enabling 5×5 coherence-weighted ROI masking (**Agent-GhibliLock ROI-Masked**) locks active foreground emblems and rails ($\text{SSIM}_{\text{edge}} = 0.9977, \rho_{\text{grad}} = 0.8899$) while leaving non-ROI background regions in smooth Ghibli gouache brushwork, further improving post-export color fidelity to $\widehat{W}_2^{\text{out}} = 1.75$ ($\widehat{SW}_2^{\text{eval}} = 1.87$) and suppressing background high-frequency sensor grain by $> 55\%$ relative to global locking.
- **Haar DWT Depth Ablation ($K \in \{1, 2, 3\}$) and Stopping Index Distribution:** Table II reports the multi-level wavelet depth ablation across $n = 60$ trials: $K = 1$ achieves the lowest $\widehat{W}_2^{\text{out}} = 1.7012$ but lower full-band $\text{SSIM}_{\text{full}} = 0.8147$; $K = 2$ (our default) balances

TABLE II
PER-STRATUM AGENT-GHIBLILOCK PERFORMANCE ($N = 600$) AND MULTI-LEVEL HAAR DWT DEPTH ABLATION ($K \in \{1, 2, 3\}$).

Stratum / Ablation	n	$\text{SSIM}_{\text{edge}}$	$\widehat{\text{W}}_2^{\text{StepA}}$	$\widehat{\text{W}}_2^{\text{StepB}}$	$\widehat{\text{W}}_2^{\text{out}}$	$\widehat{\text{SW}}_2^{\text{eval}}$	Lat. (ms)
S_1: Portrait & Emblem	150	0.9982	0.8452	1.8173	1.7395	1.8577	118.4
S_2: Perspective Rails	150	0.9946	0.9042	2.3382	2.2772	2.3830	105.8
S_3: Natural Foliage	100	0.9950	0.8967	2.3170	2.2601	2.3686	117.8
S_4: Video GOP (8-fr)	100	0.9982	0.8471	1.8317	1.7516	1.8660	142.5
S_5: Adv. Provenance	100	0.9981	0.8473	1.8018	1.7254	1.8476	132.1
Haar Depth $K = 1$ ($\text{SSIM}_{\text{full}} = 0.8147$)	60	0.9971	0.8450	1.7920	1.7012	1.8210	165.3
Haar Depth $K = 2$ ($\text{SSIM}_{\text{full}} = 0.9668$)	60	0.9985	0.8450	2.0255	1.9124	2.0315	189.6
Haar Depth $K = 3$ ($\text{SSIM}_{\text{full}} = 0.9745$)	60	0.9949	0.8450	2.7130	2.5590	2.6840	108.5

$\text{SSIM}_{\text{edge}} = 0.9985$, $\text{SSIM}_{\text{full}} = 0.9668$, and $\widehat{W}_2^{\text{out}} = 1.9124$; and $K = 3$ maximizes $\text{SSIM}_{\text{full}} = 0.9745$ and $\rho_{\text{grad}} = 0.9709$ at a slightly higher $\widehat{W}_2^{\text{out}} = 2.5590$. Across the $N = 600$ benchmark, the closed-loop stopping index distribution is $\tau = 1 : 244$ (40.7%), $\tau = 2 : 37$ (6.2%), $\tau = 3 : 13$ (2.2%), and $\tau = 4 : 306$ (51.0%); under the $n = 100$ high-contrast stress suite ($\tau_{\text{struct}} = 0.92, \tau_{\text{color}} = 3.2$), the decoupled $(\tau_\gamma, \tau_\alpha)$ controller converges at $\tau = 1 : 40\%, \tau = 2 : 4\%, \tau = 4 : 56\%$.

- **Video GOP Deflickering (S_4):** Across $n = 100$ trials of 64×64, 8-frame GOP sequences, unconstrained synthesis exhibits $\sigma^2_{\text{flicker}} = 347.92$. Unwarped Pixel-Space EMA reduces flicker only to 5.9002 while blurring moving edges ($\text{SSIM}_{\text{edge}} = 0.0870$), and Global Luminance Histogram Locking eliminates mean flicker ($\sigma^2_{\text{flicker}} = 0.0002$) but fails to restore spatial edges ($\text{SSIM}_{\text{edge}} = 0.0871$). Agent-GhibliLock's motion-compensated wavelet EMA achieves $\sigma^2_{\text{flicker,pre-clamp}} = 0.0000$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$, a 91.0% reduction in local LL_K tile variance ($0.2892 \rightarrow 0.0259$), and $\text{SSIM}_{\text{edge}} = 0.9982$.

C. Live Multi-Model Vertex AI Evaluation and Disaggregated Adversarial Provenance (S_5)

Table III reports the live Vertex AI Stage-0 VLM evaluation across `gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` alongside the 5 synthetic drift curriculum tiers (C1–C5). Note that $\text{PSNR}_{\text{QIM}} \approx 47.09$–$47.42$ dB specifically measures Stage-3 QIM watermark embedding fidelity between the post-Step-B stylized image $\widehat{I}$ and the final watermarked output I_{out}, while compiler latency (354.7–432.5 ms at 128×128) is reported separately from live Vertex AI network API latency (1696.1–3164.0 ms).

Table IV disaggregates the S_5 ($n = 100$) adversarial provenance results across A_1 (unstripped 2×2 pixel tamper), A_2 (pure C2PA `caBX` chunk stripping), A_{3a} (strip + ± 2 LSB noise), and A_{3b} (strip + actual PIL lossy JPEG $Q = 85$ recompression):

- **Export Self-Noise Accounting:** Prior to any external channel attack, uint8 rounding and sRGB clamping introduce an empirical DCT self-noise of $\text{RMS}(\eta^{\text{export}}) = 0.2854$ ($\sigma^2_{\text{round+clamp}} = 0.0821$, maximum observed $|\eta^{\text{export}}| = 1.2725 < \Delta_{\text{QIM}}/4 = 3.5$), yielding a pre-attack single-coefficient error rate of $p_e^{\text{export}} = 0.0000$.
- **Disaggregated ± 2 LSB vs. Lossy JPEG $Q = 85$ and Blind Coset Localization:** Under A_1, both informed and *blind majority-voted coset localization* (`expected_bits=None`) localize the exact tampered 8×8 block $(2, 3)$ on 100.0% (25/25) of trials. Under A_2, **B4** (metadata-only C2PA) drops to 0.0%, and classical additive Cox spread-spectrum DCT [20] (originally designed for informed/non-blind host subtraction, evaluated here in the blind regime) achieves only 74.34% bit accuracy due to Host-Signal Interference. Under A_{3a} (± 2 LSB) and A_{3b} (JPEG $Q = 85$), Agent-GhibliLock experiences pre-voting single-coefficient flip rates of $p_e = 0.08\%$ and $p_e = 0.45\%$, respectively, which are completely corrected by multi-block repetition voting (100.0% bit accuracy and 100.0% exact 64-bit payload recovery across $R = 4$, $R = 5$ odd voting at 64×64, and $R = 16$ at 128×128).

D. Real Natural Photographic Evaluation and Resolution Scaling (64^2 to 512^2)

To validate Agent-GhibliLock on complex natural photographs containing broadband textures, organic lighting gradients, skin tones, and fine metallic/embroidered insignia (beyond synthetic piece-wise smooth test scenes), Table V evaluates `grace_hopper.jpg` across four spatial resolutions (64×64 to 512×512, scaling QIM voting redundancy from $R = 4$ to $R = 256$ votes/bit) and `Minduka_Present_Blue_Pack.png` at 128×128 (visualized in Fig. 1g–i). Across all five real-photograph configurations, **Agent-GhibliLock (Global)** elevates $\text{SSIM}_{\text{edge}}$ from 0.2392–0.3937 (**B1**) and 0.4299–0.5661 (**B7 ControlNet**) to 0.9929–0.9966, achieves non-circular $\text{SSIM}_{\text{full}} = 0.9529$–$0.9712$ and $\rho_{\text{grad}} = 0.9365$–$0.9703$, contracts $\widehat{W}_2^{\text{out}}$ to 1.1108–2.3072, and maintains $\text{BER} = 0.0000$ under both clean PNG export and lossy JPEG $Q = 85$ recompression. **Agent-GhibliLock (ROI-Masked)** further lowers $\widehat{W}_2^{\text{out}}$ to 0.9906–2.0677 by restricting high-frequency detail locking to high-coherence portrait contours and insignia.

IV. LIMITATIONS, OPERATIONAL SCOPE, AND CONCLUSION

A. Limitations and Operational Failure Modes

Three operational boundary conditions govern production deployments of Agent-GhibliLock:

1) **Upstream Macro-Registration Requirement:** Because 2D Haar wavelet detail subbands (LH_k, HL_k, HH_k) are spatially localized within $2^k \times 2^k$ dyadic tiles, global detail locking requires that the upstream diffusion candidate I_{cand} be macro-spatially co-registered with I_{src} (e.g., via image-conditioned SDEdit, ControlNet, or optical-flow pre-warping). If I_{cand} synthesis hallucinates a completely different camera pose while global $\gamma_\tau \rightarrow 1$, ghost contours can appear; using Spatially Masked ROI Wavelet Locking (M_{ROI}) restricts detail injection strictly to registered foreground emblems and text.
2) **Megapixel CPU Latency vs. Vectorized/GPU Execution:** Because Python loops over 8×8 DCT blocks and

TABLE III

PART 1: LIVE GOOGLE CLOUD VERTEX AI STAGE-0 DECOMPOSER EVALUATION (`PROJECT="<GCP-PROJECT-ID>"`, 128×128). PART 2: CONTROLLED STRUCTURAL DRIFT CURRICULUM TIERS (C1–C5). $\mathrm{PSNR}_{\mathrm{QIM}}$ MEASURES STAGE-3 QIM WATERMARK DISTORTION RELATIVE TO POST-STEP-B $\widehat{I}$.

Evaluation Track	Model / Curriculum Tier	API Lat. (ms)	Tokens (P/C)	$\mathrm{SSIM}_{\mathrm{edge}}^{\mathrm{pre}}$	$\mathrm{SSIM}_{\mathrm{edge}}^{\mathrm{post}}$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\mathrm{StepA}}$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\mathrm{out}}$	$\widehat{\mathbf{SW}}_{\mathbf{2}}^{\mathrm{eval}}$	$\mathrm{PSNR}_{\mathrm{QIM}}$	CPU Lat. (ms)
Part 1: Live Vertex AI	`gemini-2.5-flash` (GA)	1696.09	105/65	0.1150	0.9742	0.7367	1.4421 ± 0.06	1.6016	47.36 dB	354.66
	`gemini-2.5-pro` (GA)	3163.96	105/11	0.1444	0.9778	0.6877	1.3800 ± 0.06	1.5357	47.39 dB	363.34
	`gemini-3-flash-preview`	2100.63	105/80	0.1843	0.9802	0.6456	1.3563 ± 0.04	1.5139	47.33 dB	363.96
Part 2: Drift Curriculum	Tier C1 ($\eta_{\mathrm{struct}} = 0.04$)	—	—	0.1199	0.9796	0.7443	1.3926	1.5381	47.42 dB	383.89
	Tier C2 ($\eta_{\mathrm{struct}} = 0.08$)	—	—	0.1981	0.9836	0.6527	1.3435	1.4938	47.39 dB	432.51
	Tier C3 ($\eta_{\mathrm{struct}} = 0.12$)	—	—	0.2707	0.9856	0.5013	1.2514	1.4246	47.09 dB	374.07
	Tier C4 ($\eta_{\mathrm{struct}} = 0.16$)	—	—	0.3418	0.9876	0.4182	1.1877	1.3725	47.14 dB	384.34
	Tier C5 ($\eta_{\mathrm{struct}} = 0.20$)	—	—	0.4325	0.9908	0.4055	1.1536	1.3383	47.34 dB	398.91

TABLE IV

DISAGGREGATED STRATUM S_5 ADVERSARIAL PROVENANCE MATRIX (`BENCHMARK_RESULTS_N600.JSON`). COX (1997) [20] IS EVALUATED IN THE BLIND REGIME (WITHOUT HOST SUBTRACTION) TO QUANTIFY HOST-SIGNAL INTERFERENCE.

Adversarial Regime	B4: C2PA	Blind Cox SS	Pre-Vote p_c	GhibliLock ($R = 4/5$)	GhibliLock ($R = 16$)
A_1: Unstripped 2×2 Tamper ($n = 25$)	100.0%	0.0%	—	100.0% (Blind: 100%)	100.0%
A_2: Pure `caBX` Chunk Strip ($n = 50$)	0.0%	74.34%	0.00%	100.0% (BER = 0)	100.0%
A_{3b}: Strip + ± 2 LSB Noise ($n = 25$)	0.0%	74.12%	0.08%	100.0% / 100.0%	100.0%
A_{3c}: Strip + Lossy JPEG $Q = 85$ ($n = 25$)	0.0%	71.88%	0.45%	100.0% / 100.0%	100.0%

$O(HW \log(HW))$ 1D sorting scale linearly with pixel count (95 ms at 64^2, 457 ms at 128^2, 1.7 s at 256^2, and 6.0 s at 512^2), real-time 1024×1024 compilation requires batched C++/SIMD or CUDA FFT/DCT kernels.

3) **Geometric Desynchronization of Block-Aligned 8×8 DCT QIM:** While our mid-band 8×8 DCT QIM watermark achieves 100% bit recovery and blind coset tamper localization under metadata stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression, non-grid-aligned cropping, aspect-ratio rescaling, or rotation desynchronize the 8×8 block lattice and require template resynchronization or neural geometric-invariant watermarks [10], [19].

B. Conclusion

Agent-GhibliLock resolves *The Captioning Conundrum* in cross-modal VLM-to-Diffusion pipelines by unifying spatial-mean-preserving 2D Haar wavelet detail locking (in both global and coherence-masked ROI modes), 3D CIELAB Sliced-Wasserstein optimal transport with exact orthonormal-frame perturbation bounds, motion-compensated video GOP wavelet EMA deflickering, and dual-layer ISO/IEC 19566-5 JUMBF (`caBX`) + blind mid-band 2D-DCT QIM provenance compilation.

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TABLE V

REAL NATURAL PHOTOGRAPHIC BENCHMARK AND RESOLUTION SCALING (64×64 TO 512×512) ON GRACE_HOPPER.JPG AND MINDUKA_PRESENT_BLUE_PACK.PNG (BENCHMARK_RESULTS_N600.JSON).

Real Photograph Asset	**Method / Paradigm**	**Votes** R	$\mathbf{SSIM}_{\text{edge}} \uparrow$	$\mathbf{SSIM}_{\text{full}} \uparrow$	$\boldsymbol{\rho}_{\text{grad}} \uparrow$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\text{StepA}} \downarrow$	$\widehat{\mathbf{W}}_{\mathbf{2}}^{\text{out}} \downarrow$	$\widehat{\mathbf{SW}}_{\mathbf{2}}^{\text{eval}} \downarrow$	**JPEG85 BER**	**CPU Lat. (ms)**
grace_hopper (64×64)	**B1**: Stylized Candidate	—	0.2392	0.4440	0.5482	2.7708	2.7708	2.8910	—	—
	B7: ControlNet 8×-VAE	—	0.4936	0.5968	0.7600	0.8082	1.8600	1.9420	—	—
	Agent-GhibliLock (Global)	4	**0.9951**	**0.9532**	**0.9454**	0.8082	2.3072	2.3132	**0.0000**	94.7
	Agent-GhibliLock (ROI-Masked)	4	0.9845	0.7986	0.8923	0.8082	**2.0677**	**2.1096**	**0.0000**	97.0
grace_hopper (128×128)	**B1**: Stylized Candidate	—	0.3653	0.5984	0.6432	2.8651	2.8651	2.9840	—	—
	B7: ControlNet 8×-VAE	—	0.5489	0.6912	0.7659	0.7208	1.6743	1.7810	—	—
	Agent-GhibliLock (Global)	16	**0.9961**	**0.9604**	**0.9403**	0.7208	1.5889	1.6969	**0.0000**	812.9
	Agent-GhibliLock (ROI-Masked)	16	0.9881	0.8185	0.8893	0.7208	**1.4451**	**1.5824**	**0.0000**	457.3
grace_hopper (256×256)	**Agent-GhibliLock (Global)**	64	**0.9966**	**0.9661**	**0.9398**	0.6529	1.2859	1.3948	**0.0000**	1847.1
	Agent-GhibliLock (ROI-Masked)	64	0.9908	0.8287	0.8978	0.6529	**1.1854**	**1.3112**	**0.0000**	1703.1
grace_hopper (512×512)	**Agent-GhibliLock (Global)**	256	**0.9944**	**0.9712**	**0.9365**	0.6374	1.1108	1.2259	**0.0000**	6993.0
	Agent-GhibliLock (ROI-Masked)	256	0.9878	0.7656	0.8851	0.6374	**0.9906**	**1.1188**	**0.0000**	6013.2
minduka_pack (128×128)	**Agent-GhibliLock (Global)**	16	**0.9929**	**0.9529**	**0.9703**	0.5575	1.6623	1.7677	**0.0000**	294.8
	Agent-GhibliLock (ROI-Masked)	16	0.9885	0.8053	0.9118	0.5575	**1.5045**	**1.5942**	**0.0000**	298.5